

Cybersecurity in Healthcare

A survey of
cybersecurity
vulnerabilities in
healthcare systems

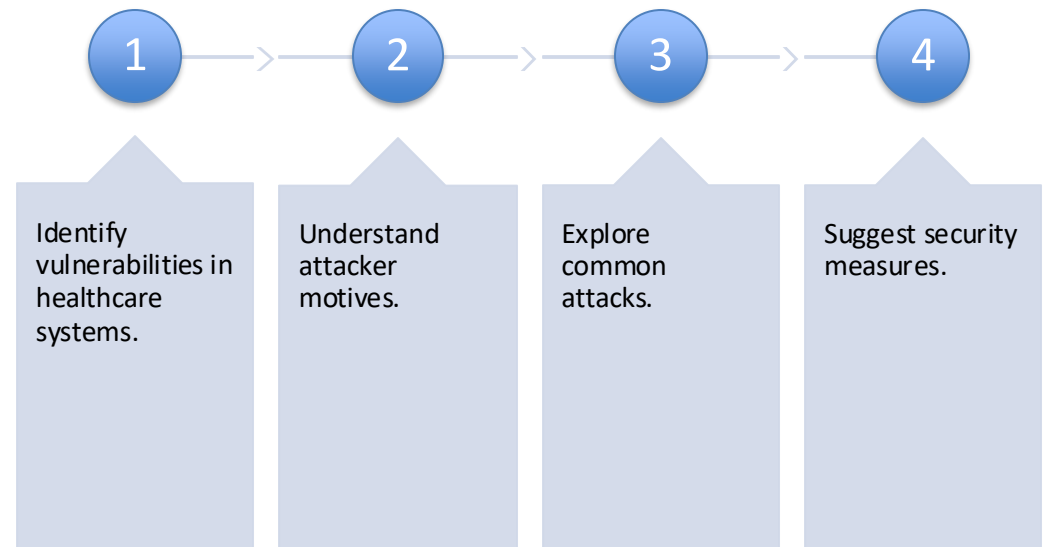
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Introduction

- Healthcare systems rely on IT and networks.
 - Cyberattacks increased during COVID-19.
 - Goal: Protect patient data confidentiality, integrity, availability.
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Key Objectives

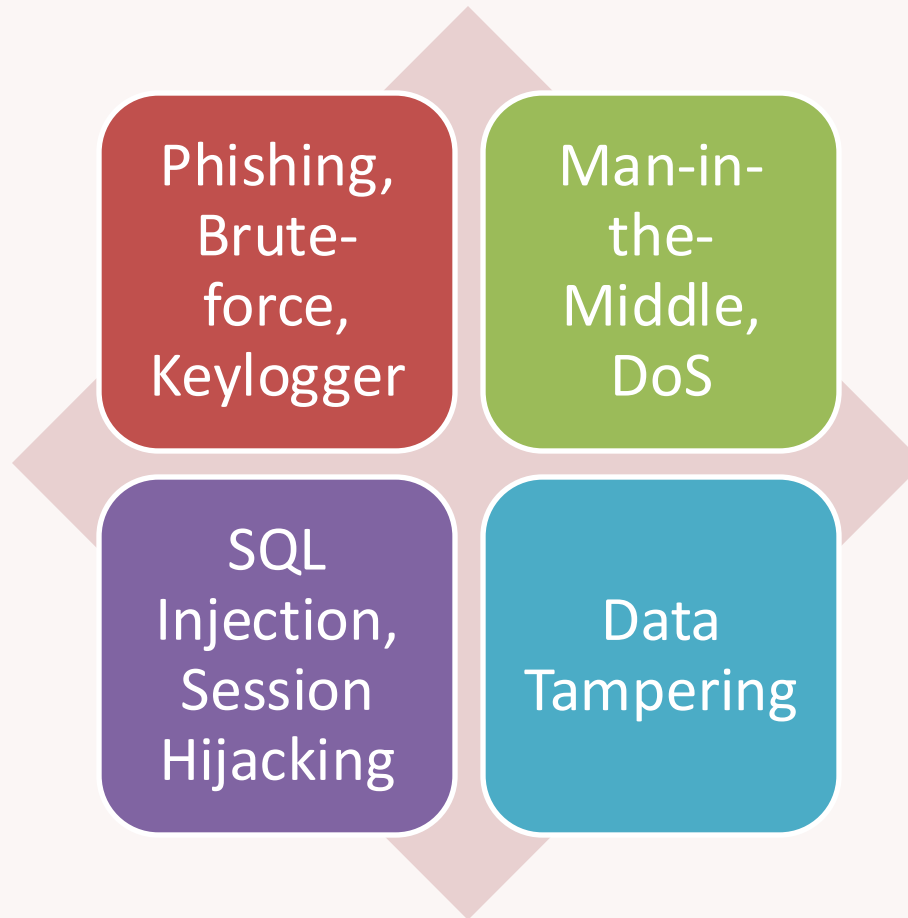


Types of Attackers

- Passive: Eavesdropping, no system disruption.
- Active: Modify data, disrupt systems, impersonate devices.



Common Cyber Attacks



Impact of Attacks

- Data breaches
- Loss of patient trust
- System downtime
- Threats to patient safety



Attack Approaches



Attack Method (Passive/Active)



Attack Complexity



Privilege Requirement



User Collaboration

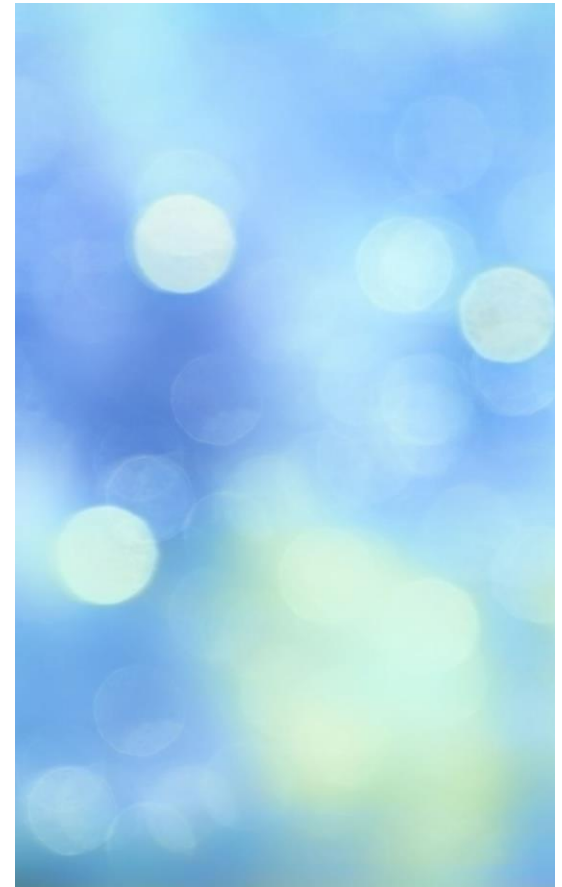
Challenges in Healthcare Security

Resistance to change

Integration issues

Large data handling

Mobile app risks



Security Requirements (CIA)

- Confidentiality: Protect patient data
- Integrity: Prevent data tampering
- Availability: Ensure access to data



System-Level Security

- Sensor level: Lightweight security
- Server level: Authentication & encryption
- Access control & key management



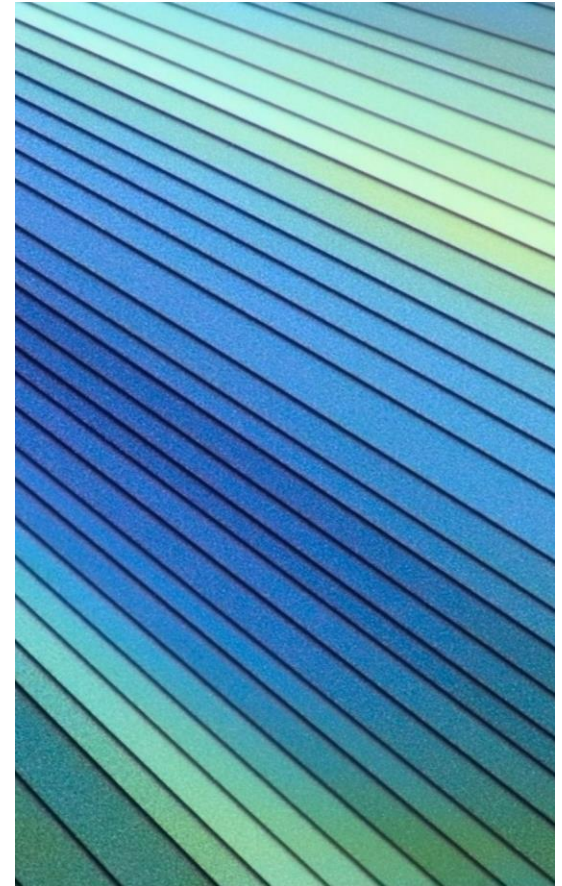
Best Practices

Intrusion Detection Systems

VPN implementation

AI-based detection

Secure communication protocols



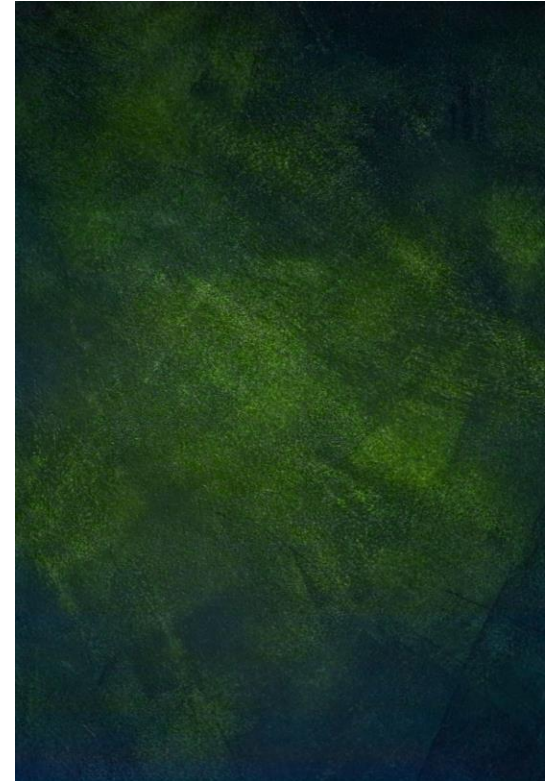
Security Issues

Unencrypted data transmission

Unauthorized access

Device vulnerabilities

Human errors



Conclusion



Healthcare systems are high-risk targets.



Continuous monitoring is essential.



Strong security practices protect patient safety.